

AREA SCIENCE PARK
RESEARCH AND TECHNOLOGY INSTITUTE
LABORATORY OF DATA ENGINEERING

Web Application for DPCfam and DPCstruct Data Exploration

Report 1 (12 Jan - 19 Jan 2026) : Basic Data
Exploration

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1 Introduction

This report is the first of a sequence in which we describe the accomplishment of timely (weekly) assigned tasks during our internship. The main focus here is the exploration of datasets to be used in the development and deployment of our browser-based web application. Essentially, these datasets are **DPCFam**([Russo et al., 2022](#)) and **DPCStruct**([Barone et al., 2025](#)). Both were derived using adapted Density Peak Clustering algorithms([Rodriguez and Laio, 2014](#)) run on protein domain features (sequences for the former and structures for the latter). Their main content consists of **metaclusters (MCs)** and some relevant metadata. Each MC is a collection of protein regions, called seed sequence or structure regions. It is stated that sequences belonging to the same *metacluster* are likely to share a core evolutionary module, and thus metaclusters can often be assimilated to protein families([Russo et al., 2022](#)).

2 Development

2.1 DPCFam

In this section, we refer to **DPCFam** as a dataset produced by running the DPCFam algorithm([Russo et al., 2022](#)) on the **UniRef50** v.2017_07 dataset, a protein sequence dataset including approximately 23 million sequences that share no more than 50% sequence identity with each other. The training of the algorithm identified approximately 45 000 protein clusters in UniRef50, and these outputs are in close correspondence with those of the **PFam** ($\approx 81\%$ matching) and **ECOD** ($\approx 72\%$ matching) resources. Likewise, the authors identified more than 14 000 clusters of protein regions with no PFam annotation, corresponding to probable novel protein families.

Initially, the authors published a dataset (**Standard DPCfam dataset**) containing MCs with more than 50 seed sequence regions and an average sequence length greater than 50 amino acids. Later on, they extended their dataset with the DPCFam-B dataset, containing smaller metaclusters with ($25 \leq N < 50$) seed sequence regions and an average sequence length ≥ 50 amino acids.

The dataset being open access on Zenodo, after downloading it, we carefully explored the **README.md** file and successfully ran the provided bash script (**parser.sh**) on a small dataset (**DPCFamB**). This choice was made not only for its simplicity but also because our available RAM was insufficient for the larger dataset (**Standard DPCFam dataset**). As a result, we obtained two directories:

1. **mc_info_from_xml**

2. mc_info_from_xml/MC_fasta

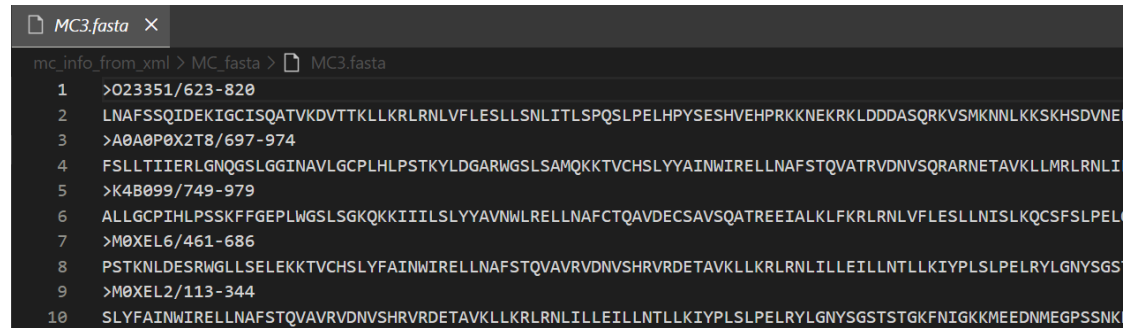
Let's describe each of them :

2.1.1 MCs fasta files

For each MC, the parser generated a FASTA file. The fasta file of MetaCluster XXX is named as **MCXXX.fasta** and contains the unaligned seed sequences of MC XXX. Each sequence is named after the original protein identifier specifying start and end position of the seed subregion in the MC, as

`>Protein_uniprot_ID/start_position-end_position.`

For instance, we have :



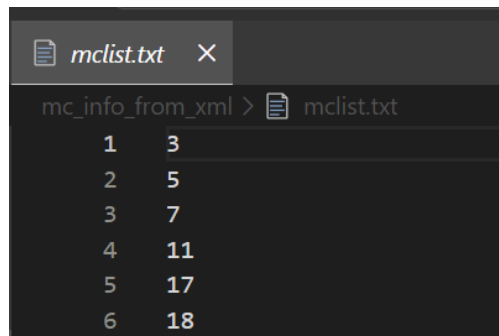
```
MC3.fasta
mc_info_from_xml > MC_fasta > MC3.fasta
1 >O23351/623-820
2 LNAFSSQIDEKIGCISQATVKDVTTKLLKRLRNLVLFLELLSNLITLSPQSLPELHPYSESHVEHPKKNEKRKLDDASQKVKSMKNLKKSKHSDVNE
3 >A0A0P0X2T8/697-974
4 FSLLTIIERLGNQGSLLGGINAVLGCPLHLPSTKYLDGARWGSLSAMQKKTVCCHSLYYAINWIRELLNAFSTQVATRVNDVNSQRARNETAVKLLMRLRLNI
5 >K4B099/749-979
6 ALLGCPILPSSKFFGEPLWGSLSGKQKKIIILSLYYAVNWLRELLNAFCTQAVDECSAVSQATREEIALKLFKRLRNLVLFLELLNISLKQCSFSLPEL
7 >M0XEL6/461-686
8 PSTKNLDESRLGLLSELEKKTVCCHSLYFAINWIRELLNAFSTQVAVRVNDVNSHRVDETAVKLLKRLRLILLEILLNTLLKIYPLSLPELRYLGNYSGS
9 >M0XEL2/113-344
10 SLYFAINWIRELLNAFSTQVAVRVNDVNSHRVDETAVKLLKRLRLILLEILLNTLLKIYPLSLPELRYLGNYSGSTSTGKFNIGKKMEEDNMEGPSSNK
```

Figure 1: Unaligned seed sequences of Metacusters.

2.1.2 SSV tables

In the folder `mc_info_from_xml` the parser generated the following SSV tables:

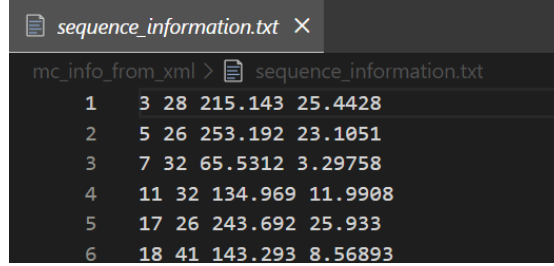
1. `mclist.txt` lists the indexes (XXX) of all MCs(**34 556**) listed in the original XML file.



```
mclist.txt
mc_info_from_xml > mclist.txt
1 3
2 5
3 7
4 11
5 17
6 18
```

Figure 2: List of metacusters.

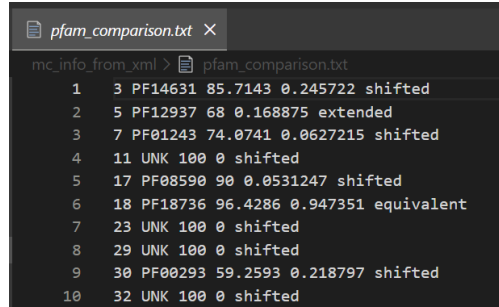
2. `sequence_information.txt` lists for each MC the following measures:



MC index	Number of sequences in the MC's seed	Average length of the MC's seed regions	Standard deviation of the length of the MC's seed regions	
1	3	28	215.143	25.4428
2	5	26	253.192	23.1051
3	7	32	65.5312	3.29758
4	11	32	134.969	11.9908
5	17	26	243.692	25.933
6	18	41	143.293	8.56893

Figure 3: Information about Sequences.

- column 1: MC index
 - column 2: Number of sequences in the MC's seed
 - column 3: Average length of the MC's seed regions
 - column 4: Standard deviation of the length of the MC's seed regions
3. `pfam_comparison.txt` lists for each MC the comparison with Pfam ground truth:



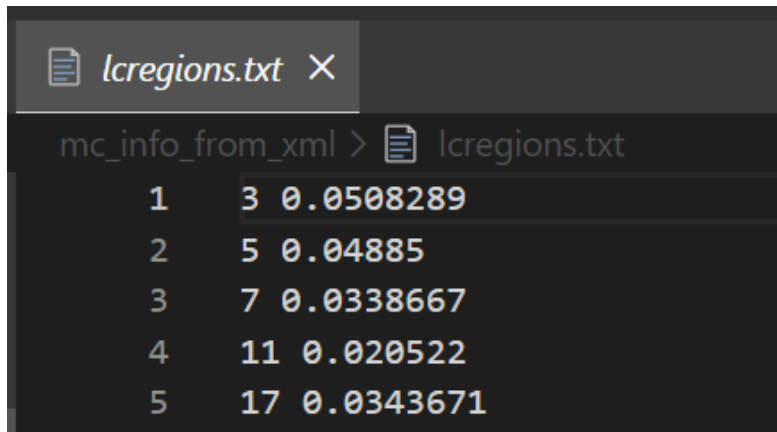
MC index	MC's Dominant Pfam Architecture (DA)	Percent of sequences in the MC matching the DA	Average overlap of the former sequences with the DA	Label
1	3 PF14631	85.7143	0.245722	shifted
2	5 PF12937	68	0.168875	extended
3	7 PF01243	74.0741	0.0627215	shifted
4	11 UNK	100	0	shifted
5	17 PF08590	90	0.0531247	shifted
6	18 PF18736	96.4286	0.947351	equivalent
7	23 UNK	100	0	shifted
8	29 UNK	100	0	shifted
9	30 PF00293	59.2593	0.218797	shifted
10	32 UNK	100	0	shifted

Figure 4: Comparison with PFam

- column 1: MC index
- column 2: MC's Dominant Pfam Architecture (DA) [UNK if no DA has been found].
- column 3: Percent of sequences in the MC matching the DA
- column 4: Average overlap of the former sequences with the DA
- column 5: Label describing qualitatively the overlap of the MC with its DA, which can be: *equivalent*, *reduced*, *extended* or *shifted*.

It is noted that in very few cases the DA of an MC is not computable (NA).

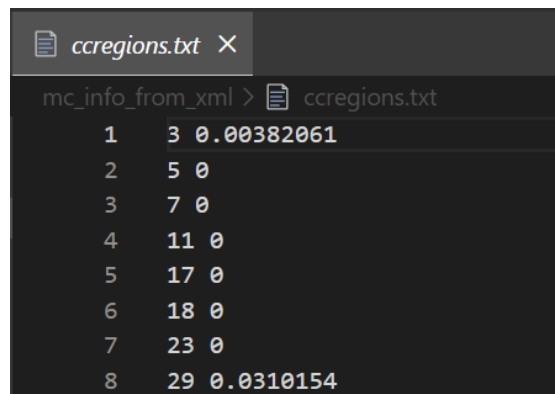
4. `lcregions.txt` lists for each MC the fraction of amino-acids in a Low Complexity region, as:



1	3	0.0508289
2	5	0.04885
3	7	0.0338667
4	11	0.020522
5	17	0.0343671

Figure 5: Low Complexity(LC) Regions.

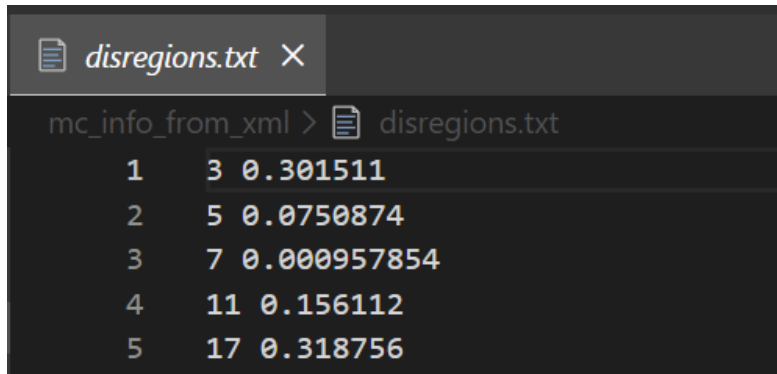
- column 1: MC index
 - column 2: LC fraction
5. `ccregions.txt` lists for each MC the fraction of the amino-acids in a Coiled Coil region, as:



1	3	0.00382061
2	5	0
3	7	0
4	11	0
5	17	0
6	18	0
7	23	0
8	29	0.0310154

Figure 6: Coiled Coil(CC) Regions.

- column 1: MC index
 - column 2: CC fraction
6. `disregions.txt` lists for each MC the fraction of amino-acids in a Disordered region, as:

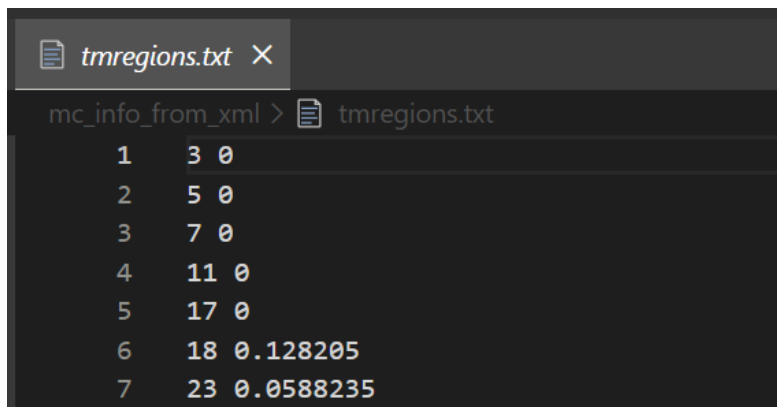


1	3	0.301511	
2	5	0.0750874	
3	7	0.000957854	
4	11	0.156112	
5	17	0.318756	

Figure 7: Disordered Regions.

- column 1: MC index
- column 2: DIS fraction

7. `tmregions.txt` lists for each MC the average number of TransMembrane regions, as:



1	3	0	
2	5	0	
3	7	0	
4	11	0	
5	17	0	
6	18	0.128205	
7	23	0.0588235	

Figure 8: TransMembrane Regions.

- column 1: MC index
- column 2: Average TM

Observation : By considering all elements given above and the `.hmm`, `.msa` files from zenodo, we retrieve everything needed to reproduce **Federico's web app** :

Name	Size Uni50	Avg. Len	% LC	% CC	% DIS	TM	Pfam DA	% DA	Size Uni50-UniKB	Overlap type
MC1	17931	185.68	4.72	0	18.44	0.01	PF13614	44.23	14317	equivalent
MC100	81	107.57	0.62	0	17.4	0	UNKOWN	100	61	None
MC100000	67	165.13	2.66	0	17.25	0.02	UNKOWN	100	51	None
MC100002	66	82.06	0.91	0.39	11.42	0	UNKOWN	100	55	None
MC100016	102	155.54	1	0.11	27.31	0.02	UNKOWN	100	83	None

Figure 9: DPCFam Web app previously developed by Federico Barone using ProcessWire content manager, using the following abbreviations :

- **Name** : Metaccluster (MC) identifier.
- **Size UniRef50** : Number of sequences belonging to the MC in UniRef50.
- **Avg. Len.** : Average length of sequences (amino acid count).
- **% LC** : Percentage of low-complexity domains.
- **% CC** : Percentage of coiled-coil domains.
- **% DIS** : Percentage of disordered domains.
- **TM** : Average number of transmembrane regions in a metaccluster.
- **Pfam DA** : Pfam architecture that best overlaps with the metaccluster. This is referred to as the dominant architecture (DA). If no overlap exists, the metaccluster is tagged as unknown (UNK).
- **% DA** : Percentage of sequences from the metaccluster in common with the dominant architecture (computed considering sequences in the UniRef50 and UniProtKB intersection).
- **Size Uni50–UniKB inter** : Number of sequences belonging to the metaccluster in the UniRef50–UniProtKB intersection.
- **Overlap type** : Classification of the overlap between the metaccluster and the dominant architecture based on how well their sequence boundaries align.

2.2 DPCStruct

In this section, we present **DPCStruct** as a dataset obtained by running the **DPCStruct algorithm**, developed by [Barone et al. \(2025\)](#), on a dataset comprising roughly 15 million proteins, representative of the **Foldseek** Clusters dataset. This dataset is a fully clustered version of the **AlphaFold** dataset, which contains roughly 214 million predicted protein structures. The final output consisted of 1 642 061 domains distributed over 1 252 179 proteins, organized into 28 246 metaclusters based on their structural similarity. We downloaded the data from Zenodo, and hereafter we present a list of its files:

1. `dpcstruct_classification.tsv`: Domains identified and their corresponding metacluster. The columns are Metacluster ID, Protein Uniprot ID, domain start, domain end as shown in Figure 10a:

	mcID	proteinID	start	end
0	0	A0A1Q3ZAL7	4	118
1	0	A0A3P1XLE5	3	122
2	0	A0A2V5NFM0	9	100
3	0	A0A6P1ZDY0	24	124
4	0	A0A6P1ZFS3	1	87

(a) Columns

Overview	
Overview	Alerts
Dataset statistics	
Number of variables	4
Number of observations	1642061
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	50.1 MB
Average record size in memory	32.0 B

(b) Profiling Report

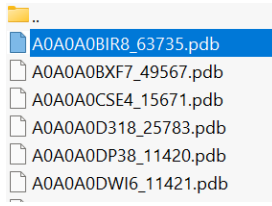
Figure 10: Domains identified per Metacluster on 1 642 061 samples. [Notebook](#).

2. `mcs_reps.fasta`: For each metacluster, two representative domains were selected: One representing the center of the cluster and the other being the domain with the highest pLDDT score. If these are the same, only one domain is included as the representative.

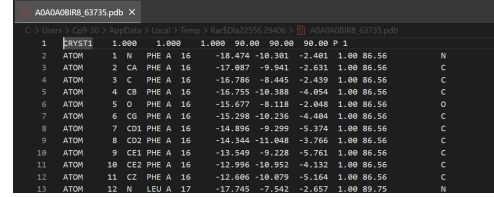
```
>A0A009IR62_1789 | 1-186
MSLMKSVVLIIFASLAVNIAYSAETNPISIQNYWSIAEQKKLDQDITWQRLMYANKNQKSEVTYAGYFLSENGKNLKEELKADISALFIPTQDNQSIRCKFPARQWLIIQQLGIQENLPQVKCSEI
>A0A009JJQ9_20571 | 7-90
IPLIDKLVKEKNFVLLTWDARYSSGAWACCLPYLNQCEVVEASEDGDTLIPKMEYLLNTNWLPIMDGSSAMDALEKLETRLA
>A0A009K0I0_6579 | 61-148
MDKQFKQGLKKASAENSPFSALGYILTTGANWAKPIENFKLTERDKNELVSFCWDGVPKKISPTQFQIKTKFVPKKDLDIIFVRVK
>A0A009QKA3_27491 | 1-105
MKPEQFIREFGEKKARALLAQHLNLGCPDDMKITVINGMWHRTSKGFTYPELKRLLSIDQINAFDGGIKEAKEILSRINKHGSKYATLFRPALEQAIHDYESI
>A0A010R0N5_15978 | 9-507
LTAEQTKALFNILTHFETYHEIEDFKPETISDYGPFAQSSKPGETVTYAPESSAPLLQSLFTRFILKLPGVSSFPVPEFWNVVRVQGIKNFAEAELESSEYKALGTRKTLATASSTVIETVARC
>A0A010RPH6_14093 | 14-358
```

Figure 11: Representative Sequence(of amino acids) per metacluster.

3. `mcs_reps_pdbs.zip`: Contains a PDB file for each representative domain. The filename is structured as `'proteinID_metacluster.pdb'`.



(a) PDB files



(b) Content of A0A0A0BIR8_63735.pdb

Figure 12: PDB files for representative domains.

4. `mcs_properties.tsv`: Set of properties per metacluster as presented in Figure 13a and explained afterwards:

mcID	size	len_aa	len_std	len_ratio	plddt	disorder	alntmscore	tmscore	lddt	prob	pident
0	61388	30643	210.005	38.0453	0.181164	82.6220	0.284676	0.2305	0.2478	0.3321	10.7832
1	49953	28922	282.901	60.4128	0.213547	81.7385	0.225913	0.3476	0.3795	0.3599	10.5943
2	52595	26312	122.438	21.1918	0.173082	84.0467	0.286809	0.3575	0.3819	0.3671	10.8535
3	42223	17503	260.899	60.0432	0.230140	78.8912	0.223797	0.4164	0.4628	0.4638	16.7721
4	124	16132	191.621	37.0314	0.193253	85.6541	0.197148	0.3631	0.3936	0.3936	12.6375

(a) Columns

Overview	
Overview	Alerts
Dataset statistics	
Number of variables	12
Number of observations	28246
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	28.10B
Average record size in memory	901B

(b) Report

Figure 13: MCs Properties : 28 246 samples. Notebook available [here](#).

- `mcID`: Metacluster ID.
- `size`: Number of domains.
- `len_aa`: Average length of domains (number of amino acids).
- `len_std`: Standard deviation of domain lengths.
- `len_ratio`: Ratio of `len_std` to `len_aa`.
- `plddt`: Average predicted LDDT as reported by AlphaFold2.
- `disorder`: Average intrinsic disorder score calculated with AIUPred.
- `alntmscore`: Pairwise alignment TM-score between domains, averaged over all pairs.
- `tmscore`: Pairwise alignment TM-score between domains, averaged over all pairs, using the maximum between TM-score normalized by query or target.
- `lddt`: Pairwise LDDT score, averaged over all pairs.
- `prob`: Pairwise probability of homology according to SCOPE, as reported by Foldseek.
- `pident`: Pairwise percentage identity, averaged over all pairs.

5. annotated_[cath|scop]_qc[x]_t[x]_l[x]:

```

├── annotated_cath_qc0.5_t0.5_l0.5.tsv
├── annotated_cath_qc0.8_t0.5_l0.5.tsv
├── annotated_scop_qc0.5_t0.5_l0.5.tsv
└── annotated_scop_qc0.8_t0.5_l0.5.tsv

```

Figure 14: Tree of annotated_[cath|scop]_qc[x]_t[x]_l[x]

For each folder in [CATH|SCOP], we are provided with the best matching DPCstruct domain, if available, along with the structural alignment information as reported by Foldseek. A fold is considered annotated if its alignment values meet or exceed the following thresholds:

- qc: query coverage.
- t: template modelling score of the alignment.
- l: lddt score of the alignment.

Basically, each subfolder(cath or scop) contains two files, let's have a quick look into them:

5.1 annotated_cath_qc0.5_t0.5_l0.5.tsv

	query	target	qstart	qend	tstart	tend	qlen	qlen	qcov	tcov	alnlen	pident	eval	bits	qtscore	ttmscore	alntmscore	lddt	prob
0	4.10.270	A0A7F8Q518_39405	1	55	127	181	68	182	0.809	0.302	55	61.8	1.218000e-02	88	0.5797	0.2543	0.2543	8.906E-01	0.995
1	2.170.210	A0A7J6I9T2_8385	25	168	35	172	172	180	0.837	0.767	156	12.1	4.788000e-01	37	0.2210	0.2132	0.2132	5.044E-01	0.164
2	3.40.20	A0A416P6C2_7370	7	134	27	155	134	160	0.955	0.806	151	11.9	7.372000e-01	-2147483648	0.1982	0.1710	0.1710	-NAN	0.000
3	3.90.105	A0A126G1X7_6894	2	58	96	160	58	281	0.983	0.231	69	18.8	6.116000e-03	120	0.8315	0.1831	0.1831	8.005E-01	1.000
4	1.20.1740	A0A2J6WLT6_15600	11	417	2	411	435	412	0.936	0.995	427	21.7	3.007000e-15	682	0.8248	0.8685	0.8685	6.804E-01	1.000

Figure 15: 1311 samples in 19 columns with 95 missing cells (0.4%) in lddt.

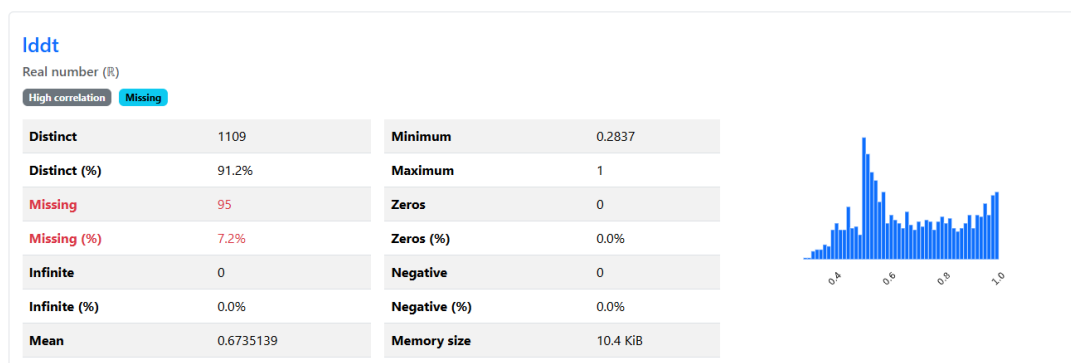


Figure 16: Report lldt column : 7.2% missing values. Notebook.

5.2 annotated_cath_qc0.8_t0.5_10.5.tsv

Same variables as in Figure 15, 1274 samples, 116 missing cells (0.5% of the whole dataset, but 9.1% in the column) in lddt. [Notebook](#).

5.3 annotated_scop_qc0.5_t0.5_10.5.tsv

	query	target	qstart	qend	tstart	tend	qlen	tlen	qcov	tcov	alnlen	pident	evalue	bits	qtmcscore	ttmscore	alntmscore	lddt	prob
0	R4YEY9_2000000	A0A6N8Q492_4003	1	49	208	256	49	270	1.000	0.181	49	97.9	6.981000e-07	246	0.9979	0.1814	0.1814	1.0000	1.0
1	P39805_2000001	R6QN42_48898	1	54	1	54	55	68	0.982	0.794	54	50.0	7.142000e-07	303	0.9598	0.7799	0.7799	0.9533	1.0
2	USKQB1_2000002	A0A1K0FU75_8415	1	134	29	162	134	163	1.000	0.822	134	87.3	2.588000e-17	857	0.9990	0.8214	0.8214	0.9998	1.0
3	Q9X1H2_2000003	A0A7C2KEC5_36025	1	186	1	186	203	186	0.916	1.000	186	97.8	5.614000e-36	1563	0.9157	0.9993	0.9993	0.9524	1.0
4	Q87WW1_2000004	A0A0P9IV70_61195	1	105	105	209	105	213	1.000	0.493	105	99.0	1.595000e-18	874	0.9978	0.4924	0.4924	0.9967	1.0

Figure 17: 1 411 samples in 19 columns, no missing, no duplicates. [Notebook](#).

5.4 annotated_scop_qc0.8_t0.5_10.5.tsv

Same variables as in Figure 17, 1359 samples, no missing, no duplicates. [Notebook](#).

6. dpcstruct_consistency.tsv: Consistency of DPCstruct metaclusters with respect to Pfam 36.0 labels.

	mcID	score	pfam-labels
0	5	1.00	CL0263
1	6	1.00	PF14802
2	7	1.00	PF00110
3	8	1.00	CL0465
4	9	0.75	CL0272-CL0016

(a) Columns

Overview

Overview	Alerts 1	Reproduction
Dataset statistics		
Number of variables	3	
Number of observations	14423	
Missing cells	0	
Missing cells (%)	0.0%	
Duplicate rows	0	
Duplicate rows (%)	0.0%	
Total size in memory	338.2 KiB	
Average record size in memory	24.0 B	

(b) Overview

Figure 18: 14423 samples across 3 columns, 7539 distinct labels. [Notebook](#).

The authors ([Barone et al., 2025](#)) considered a Pfam label to overlap with a DPCstruct domain even if it shares just one amino acid, which is why some metaclusters have many labels. In such cases, we only display 5 representative labels.

7. pfam_consistency.tsv: Consistency of Pfam Clans with respect to DPCstruct labels.

	PfamID	score	DPCstruct-label
0	CL0001	0.577698	56862-55086-22463-4358-47813
1	CL0003	0.070468	24800-36176-37984-275-4452
2	CL0004	0.566181	32775-2976-21811-55086-61848
3	CL0005	0.735225	58948-4452-35673-61388-10731
4	CL0007	0.686627	37573-45182-49202-6006-19588

(a) Columns

Overview

Overview	Alerts 2	Reproduction
Dataset statistics		
Number of variables	3	
Number of observations	5563	
Missing cells	0	
Missing cells (%)	0.0%	
Duplicate rows	0	
Duplicate rows (%)	0.0%	
Total size in memory	130.5 KiB	
Average record size in memory	24.0 B	

(b) Overview

Figure 19: 5563 samples across 3 columns. [Notebook](#).

3 Conclusion

Within this first draft, we partially explored the DPCFam and DPCStruct datasets. So far, we have identified all the necessary components to reproduce the existing DPCFam web application. In the coming days, reaching an agreement on how to properly organize and store the data described above, along with the use of Orfeo, could lead us to a consistent dataset, opening the door to the design and development of our unified web application.

References

- Barone, F., Laio, A., Punta, M., Cozzini, S., Ansuini, A., and Cazzaniga, A. (2025). Unsupervised domain classification of alphafold2-predicted protein structures. *PRX Life*, 3(2):023009.
- Rodriguez, A. and Laio, A. (2014). Clustering by fast search and find of density peaks. *science*, 344(6191):1492–1496.
- Russo, E. T., Barone, F., Bateman, A., Cozzini, S., Punta, M., and Laio, A. (2022). Dpcfam: unsupervised protein family classification by density peak clustering of large sequence datasets. *PLOS Computational Biology*, 18(10):e1010610.